Note

If b is negative, $\frac{b}{|b|}=-1$, x and y have different signs.

If b is positive, $\frac{b}{|b|}=1$, x and y have same sign.

Example 2.17

Find the square root of $6-8i$.

Solution

We compute $|6-8i| = \sqrt{6^2+(-8)^2} = 10$

and applying the formula for square root, we get

$$\sqrt{6-8i} = \pm\left(\sqrt{\frac{10+6}{2}} - i\sqrt{\frac{10-6}{2}}\right) \qquad (\because b \text{ is negative}, \frac{b}{|b|}=-1)$$

$$= \pm\left(\sqrt{8} - i\sqrt{2}\right)$$

$$= \pm\left(2\sqrt{2} - i\sqrt{2}\right).$$

EXERCISE 2.5

1. Find the modulus of the following complex numbers
   (i) $\frac{2i}{3+4i}$ (ii) $\frac{2-i}{1+i}+\frac{1-2i}{1-i}$ (iii) $(1-i)^{10}$ (iv) $2i(3-4i)(4-3i)$.
2. For any two complex numbers z_1 and z_2, such that $|z_1|=|z_2|=1$ and $z_1z_2 \neq -1$, then show that $\frac{z_1+z_2}{1+z_1z_2}$ is a real number.
3. Which one of the points $10-8i$, $11+6i$ is closest to $1+i$.
4. If $|z|=3$, show that $7 \leq |z+6-8i| \leq 13$.
5. If $|z|=1$, show that $2 \leq |z^2-3| \leq 4$.
6. If $|z|=2$, show that $8 \leq |z+6+8i| \leq 12$.
7. If z_1, z_2, and z_3 are three complex numbers such that $|z_1|=1, |z_2|=2, |z_3|=3$ and $|z_1+z_2+z_3|=1$, show that $|9z_1z_2+4z_1z_3+z_2z_3|=6$.
8. If the area of the triangle formed by the vertices z, iz, and $z+iz$ is 50 square units, find the value of $|z|$.
9. Show that the equation $z^3+2\bar{z}=0$ has five solutions.
10. Find the square roots of (i) $4+3i$ (ii) $-6+8i$ (iii) $-5-12i$.

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